

Anh Tay Nguyen

Ph.D. Candidate in Mechanical Engineering, Northwestern University
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Education

Northwestern University <i>Ph.D. Candidate in Mechanical Engineering; Advisor: Prof. Zdeněk P. Bažant</i>	Evanston, IL, USA <i>Expected June 2027</i>
Stony Brook University (SUNY Korea Campus) <i>M.S. in Mechanical Engineering, Summa Cum Laude; Advisor: Prof. Y. Eugene Pak</i>	Incheon, South Korea <i>June 2021</i>
Stony Brook University <i>Exchange Study (Sophomore Year)</i>	Long Island, NY, USA <i>2017-2018</i>
Stony Brook University (SUNY Korea Campus) <i>B.E. in Mechanical Engineering, Summa Cum Laude</i>	Incheon, South Korea <i>June 2020</i>

Awards & Honors

Teaching Assistant Award (Honorable Mention) <i>Department of Mechanical Engineering</i>	Northwestern University <i>2025</i>
Best Poster Award <i>American Rock Mechanics Association</i>	59th U.S. Rock Mechanics/Geomechanics Symposium <i>June 2025</i>
Walter P. Murphy Fellowship <i>Full funding for first year of Ph.D. study</i>	Northwestern University <i>2022</i>
Academic Excellence and Prestige Scholarship <i>Highest GPA in the Department of Mechanical Engineering</i>	SUNY Korea <i>2019</i>
Shared Prosperity Scholarship <i>Full tuition, program fee, room and board</i>	SUNY Korea <i>2016-2020</i>

Teaching Assistance Experience

Intro to Aerospace Engineering (Instructor: Sridhar Krishnaswamy) <i>Assisted with assignments and supervised student projects.</i>	Northwestern University <i>Spring 2026</i>
Aerodynamics (Instructor: Sridhar Krishnaswamy) <i>Assisted with assignments and supervised lab sections.</i>	Northwestern University <i>Spring 2026</i>
Stress Analysis (Instructor: Sridhar Krishnaswamy) <i>Assisted with assignments, and exams; supervised lab sections and student projects.</i>	Northwestern University <i>Spring 2025</i>
Quasibrittle Fracture and Scaling (Instructor: Zdeněk P. Bažant) <i>Delivered independent lectures and review sessions during instructor's absence.</i>	Northwestern University <i>Spring 2023, 2024</i>
Engineering Analysis II (Instructor: Yonggang Huang) <i>Led recitations, review sessions and grading for undergraduate.</i>	Northwestern University <i>Winter 2023</i>

Publications

Peer-reviewed Journal Articles

- [17] Xu, H., Zhao, Y., Le, J. L., **Nguyen, A. T.**, Deodatis, G., & Bažant, Z. P. (2026). Critique of lower-margin equations in concrete design codes and of their effect on structural safety software. *Journal of Structural Engineering*. **Accepted**.
- [16] **Nguyen, A. T. (co-first author)**, Asem, P., Zhao, Y., Labuz, J., & Bažant, Z. P. (2026). A rapid permeability test of low-porosity rock based on analysis of initial pressure pulse decay: experimental validation. *Rock Mechanics and Rock Engineering*. **Accepted**.

- [15] Xu, H., Zhao, Y., **Nguyen, A. T.**, & Bažant, Z. P. (2026). Smooth lagrangian crack band model with softening sprress–sprain relation and crack width prediction. *International Journal of Non-Linear Mechanics*, 186, 105340. [Link].
- [14] Zhao, Y., **Nguyen, A. T.**, Nguyen, H. T., & Bažant, Z. P. (2026). Osmotic control of the spacing of parallel shear cracks in shale growing subcritically in geologic past. *Journal of the Mechanics and Physics of Solids*, 207, 106428. [Link].
- [13] Zhao, Y., **Nguyen, A. T.**, Chen, H., Espinosa, H., & Bažant, Z. P. (2025). Crack-parallel stress effects in soft material fracture: insights from gap tests and the microplane modeling perspective. *Journal of Applied Mechanics*, 93(2). [Link].
- [12] **Nguyen, A. T.**, Linh, N. N., Thang, N. C., Ngoc, N. H., & Vinh, L. Q. (2025). A novel weighted averaging operator for frequency analysis of nonlinear free vibrations. *International Journal of Non-Linear Mechanics*, 105057. [Link].
- [11] **Nguyen, A. T.**, Asem, P., Zhao, Y., & Bažant, Z. P. (2024). Fast permeability measurement for tight reservoir cores using only initial data of the one-chamber pressure pulse decay test. *Journal of the Mechanics and Physics of Solids*, 192, 105805. [Link].
- [10] Xu, H., **Nguyen, A. T.**, & Bažant, Z. P. (2024). Sprain energy consequences for damage localization and fracture mechanics. *Proceedings of the National Academy of Sciences*, 121(40), e2410668121. [Link].
- [9] Linh, N. N., **Nguyen, A. T.**, Thang, N. C., Anh, N. D., & Elishakoff, I. (2024). Extension of dual equivalent linearization to analysis of deterministic dynamic systems: part 2 — multi-parameter equivalent linearization. *Nonlinear Dynamics*, 112(20), 18001–18030. [Link].
- [8] Kienzler, R., **Nguyen, A. T.**, & Pak, Y. E. (2024). M–integral and energy-release rates: A didactical account. *Advanced Structured Materials*, 89–120. [Link].
- [7] **Nguyen, A. T.**, Xu, H., Matouš, K. and Bažant, Z. P. (2023). Smooth Lagrangian crack band model based on Sprress–sprain relation and Lagrange multiplier constraint of displacement gradient. *Journal of Applied Mechanics*, 91(3). [Link].
- [6] Bažant, Z. P., & **Nguyen, A. T.** (2023). Osmotic pressure gradient effects on water diffusion in porous rock: Can they pervert permeability tests? *Journal of Applied Mechanics*, 90(12). [Link].
- [5] Anh, N. D., Linh, N. N., Long, T. T., Thang, N. C., **Nguyen, A. T.**, & Elishakoff, I. (2022). Extension of dual equivalent linearization to analysis of deterministic dynamic systems. part 1: Single-parameter equivalent linearization. *Nonlinear Dynamics*, 111(2), 997–1017. [Link].
- [4] **Nguyen, A. T.**, Thang, N. C., Long, T. T., Anh, N. D., Thang, P. M., & Thanh, N. X. (2022). A novel weighted local averaging for the Galerkin method with application to elastic buckling of Euler column. *International Journal for Computational Methods in Engineering Science and Mechanics*, 24(2), 128–142. [Link].
- [3] Linh, N. N., **Nguyen, A. T.**, Manh, N. V., Tuan, V. A., Kuu, N. V., Anh, N. D., & Elishakoff, I. (2021). Efficiency of mono-stable piezoelectric Duffing energy harvester in the secondary resonances by averaging method, part 2: Super-harmonic resonance. *International Journal of Non-Linear Mechanics*, 137, 103817. [Link].
- [2] **Nguyen, A. T.**, Seo, Y., Park, S.-H., & Pak, Y. E. (2021). Quantification of the effects of misfit strain on the energy states of zinc-blende spherical core/shell quantum dots. *Journal of the Korean Physical Society*, 79(1), 87–94. [Link].
- [1] Anh, N. D., Linh, N. N., Manh, N. V., Tuan, V. A., Kuu, N. V., **Nguyen, A. T.**, & Elishakoff, I. (2020). Efficiency of mono-stable piezoelectric Duffing Energy Harvester in the secondary resonances by averaging method. part 1: Sub-harmonic resonance. *International Journal of Non-Linear Mechanics*, 126, 103537. [Link].

Book Chapter

- [1] **Nguyen, A. T.**, Garnaik, S. S., Kienzler, R., & Pak, Y. E. (2024). Path-independent integrals and their applications in fracture and defect mechanics. In *Deformation and Fracture in Materials: Advances in Experimental and Computational Mechanics*. [Link].

Conference Proceedings (Peer-Reviewed)

- [6] **Nguyen, A. T.**, Zhao, Y., Niu, Y., & Bažant, Z. P. (2026). Rapid in-situ permeability estimation of reservoir rocks from fall-off tests using a parabolic approximation of axisymmetric diffusion. *60th US Rock Mechanics/Geomechanics Symposium*, Tucson, Arizona, USA, American Rock Mechanics Association.
- [5] Xu, H., **Nguyen, A. T.**, Zhao, Y., & Bažant, Z. P. (2025). A poromechanical model for the branching of hydraulic fractures in rocks with pre-existing weak layers. *59th US Rock Mechanics/Geomechanics Symposium*, Santa Fe, New Mexico, USA, American Rock Mechanics Association. [Link].
- [4] **Nguyen, A. T.**, Asem, P., Matter, J., Labuz, J. F., & Bažant, Z. P. (2025). Fast permeability measurements of tight rock: Theoretical study and experimental validation. *59th US Rock Mechanics/Geomechanics Symposium*, Santa Fe, New Mexico, USA, American Rock Mechanics Association. [Link].
- [3] Asem, P., **Nguyen, A. T.**, Bažant, Z. P., Matter, J., & Labuz, J. F. (2024). Permeability of serpentinitized harzburgite and its implications on mineral carbon storage. *58th US Rock Mechanics/Geomechanics Symposium*, Golden, Colorado, USA, American Rock Mechanics Association. [Link].
- [2] Bažant, Z. P., **Nguyen, A. T.**, Xu, H., Asem, P., & Labuz, J. F. (2024). Hydraulic crack propagation and rock permeability under osmotic pressure gradients with implications for deep CO₂ sequestration and fracking. *58th US Rock Mechanics/Geomechanics Symposium*, Golden, Colorado, USA, American Rock Mechanics Association. [Link].
- [1] Bažant, Z. P., Xu, H., **Nguyen, A. T.**, Carey, J. W., & Khan, M. (2023). The prospect of massive sequestration of atmospheric CO₂ in deep formations of basalt or peridotite appraised by fracture, diffusion and osmosis analysis. *57th US Rock Mechanics/Geomechanics Symposium*, Atlanta, Georgia, USA, American Rock Mechanics Association. [Link].

Manuscripts In Preparation

- [1] **Nguyen, A. T.**, Zhao, Y., Chen, H., Espinosa, H. D., & Bažant, Z. P. Experimental and numerical characterization of crack-parallel stress effects on mode-I fracture of shale. (Target journal: *Journal of the Mechanics and Physics of Solids*).

Talks and Posters

- [9] **Nguyen, A. T.**, Zhao, Y., Niu, Y., & Bažant, Z. P. (2026). Rapid in-situ permeability estimation of reservoir rocks from fall-off tests using a parabolic approximation of axisymmetric diffusion. *60th US Rock Mechanics/Geomechanics Symposium*, Tucson, Arizona, USA, American Rock Mechanics Association.
- [8] **Nguyen, A. T.**, Zhao, Y., & Bažant, Z. P. (2026). A finite-strain microplane damage model for fracture of soft materials. *Engineering Mechanics Institute Conference (EMI)*, Boulder, Colorado, USA.
- [7] **Nguyen, A. T.**, Asem, P., Zhao, Y., Labuz, J. F., & Bažant, Z. P. (2026). A rapid permeability test of low-porosity rock based on analysis of initial pressure pulse decay: experimental validation. *GMCS Annual Workshop*, Minneapolis, Minnesota, USA.
- [6] **Nguyen, A. T.**, Zhao, Y., Chen, H., Espinosa, H. D., & Bažant, Z. P. (2025). Crack-parallel stress effect on fracture properties of eagle ford shale identified from gap test. Presented at *Engineering Mechanics Institute Conference (EMI)*, Anaheim, California, USA and *IMECE International Mechanical Engineering Congress*, Memphis, Tennessee, USA.
- [5] **Nguyen, A. T.**, Asem, P., & Bažant, Z. P. (2024). Fast test of rock permeability with osmotic effects: a mathematical model. *GMCS Annual Workshop*, Minneapolis, Minnesota, USA.
- [4] **Nguyen, A. T.**, Asem, P., Labuz, J. F., & Bažant, Z. P. (2024). Effect of Osmotic Pressure on Water Diffusion and Hydraulic Fracturing of Mafic Rocks Aimed at Sustainable Deep Sequestration of CO₂. *Engineering Mechanics Institute Conference (EMI)*, Chicago, Illinois, USA.
- [3] **Nguyen, A. T.**, Nguyen, H. T., & Bažant, Z. P. (2023). Osmotic Ion Concentration Control of Steady-State Subcritical Fracture Growth in Shale. *Engineering Mechanics Institute Conference (EMI)*, Atlanta, Georgia, USA.

- [2] **Nguyen, A. T.**, & Pak, Y. E. (postponed to August 22–27, 2021 due to COVID-19). Effects of Misfit Strain on Quantum Mechanical States of Spherical Core/Shell Quantum Dots [Poster]. *25th International Congress of Theoretical and Applied Mechanics (ICTAM)*, Milano, Italy.
- [1] **Nguyen, A. T.**, & Pak, Y. E. (2020, December 16–19). Elastic Behavior and Energy Release Rate of a 2-D Circular Misfitting Hollow Inclusion in a Finite Matrix. *Korean Society of Mechanical Engineers (KSME) Academic Conference*, Gangwon, South Korea.